



Inspiring Excellence

BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *IoT-Enabled Biometric Rickshaw Meter & Security System*

by

Group_No: 07

Section: 09

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Abstract

This project intends to transform the transportation in the area by coming up with the IoT-Enabled Biometric Rickshaw Meter and Security System. The system focuses on three important problems, i.e. illegal use of the vehicle, irregular fare system, and the security of the passengers. The methodology will enable a combination with an Adafruit Fingerprint Sensor to provide biometric authentication means, so that only authorized drivers can turn on the motors. The calculation of fare is done automatically with the help of Hall Effect sensor and RFID technology that helps to trace the distance and control the trip sessions. Additionally, an MPU6050 Accelerator helps to monitor the driving behavior (when a car drives rashly or crashes) and provides an emergency stop. Important findings show a secure unlocking, real time distance to fare conversion shown on an LCD and responsive bluetooth. The project also ensures sustainability through digitising the metering to reduce fraud and improving the safety of the residence by detecting accidents in the community.

Keywords

Biometric Security, Smart Metering, IoT, Embedded Systems, Rash Driving Detection.

1. Introduction

Problem & Motivation: Problematic Traditional rickshaw services have haggling over the price, no anti-theft systems of drivers and uncontrolled reckless driving. This sector lacks digital integration that will result in inefficiency and safety.

Objectives:

- To come up with a security system that opens the car only through fingerprint verification.
- To have an RFID-based trip management digital metering system based on fair and distance-based.
- To provide a heightened level of safety through the ability to detect rash driving forces and halt the vehicle automatically during a crash.

Scope & Relevance: This project lies within the scope of Computer Interfacing (CSE360), utilizing protocols such as UART, I2C, SPI, and hardware interrupts to interface sensors (Fingerprint, MPU6050, RFID) with an Arduino microcontroller.

2. Related Work/ Inspiration

Current Systems: Current digital meters on auto-rickshaws are not biometric and use simple keys which can be copied. Ride-sharing applications are also offered with GPS tracking but need costly smartphones and access to the internet, which is not always possible to all rickshaw pullers.

Differentiation: Our system is differentiated by being fully offline and based on embedded hardware. It also implements biometric ignition, that is not common in cheap cars, and also incorporates the

accelerometer-based crash detection that literally interferes with the motor power, addressing safety concerns that more basic meters disregard.

3. Technical Approach

System Architecture:

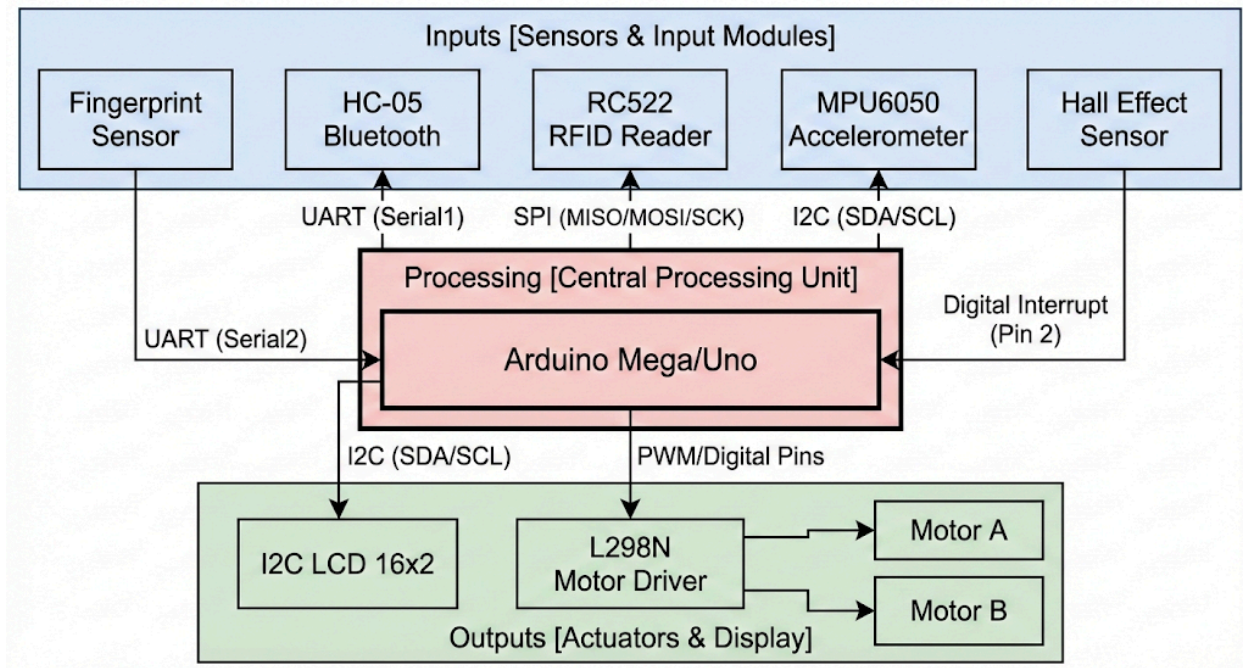


Fig: System Block Diagram

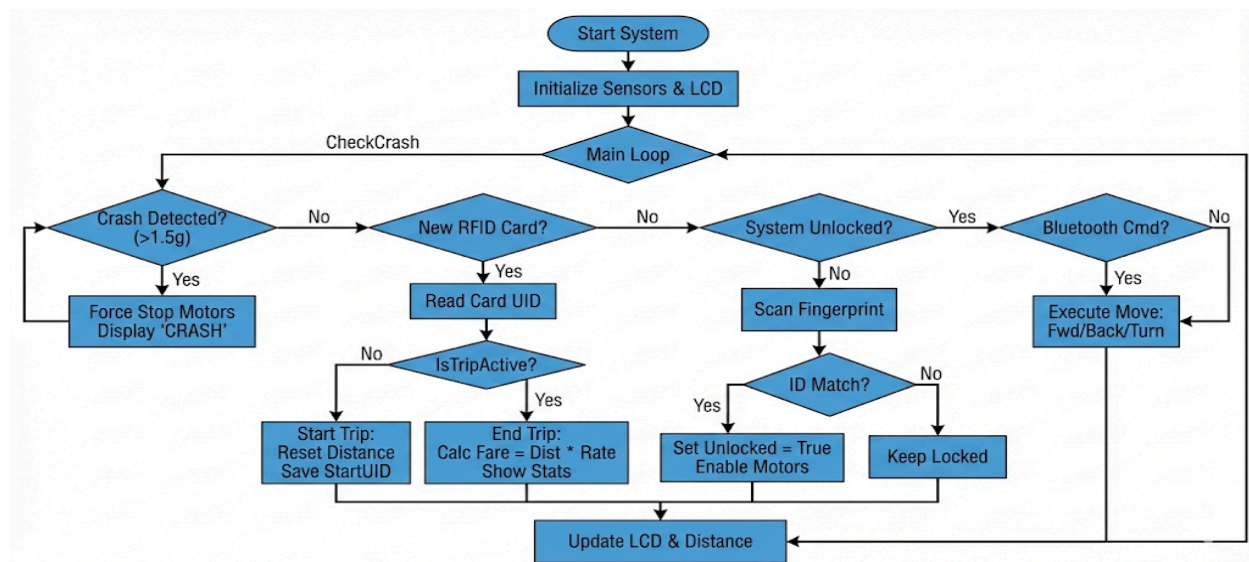


Fig: System Flowchart

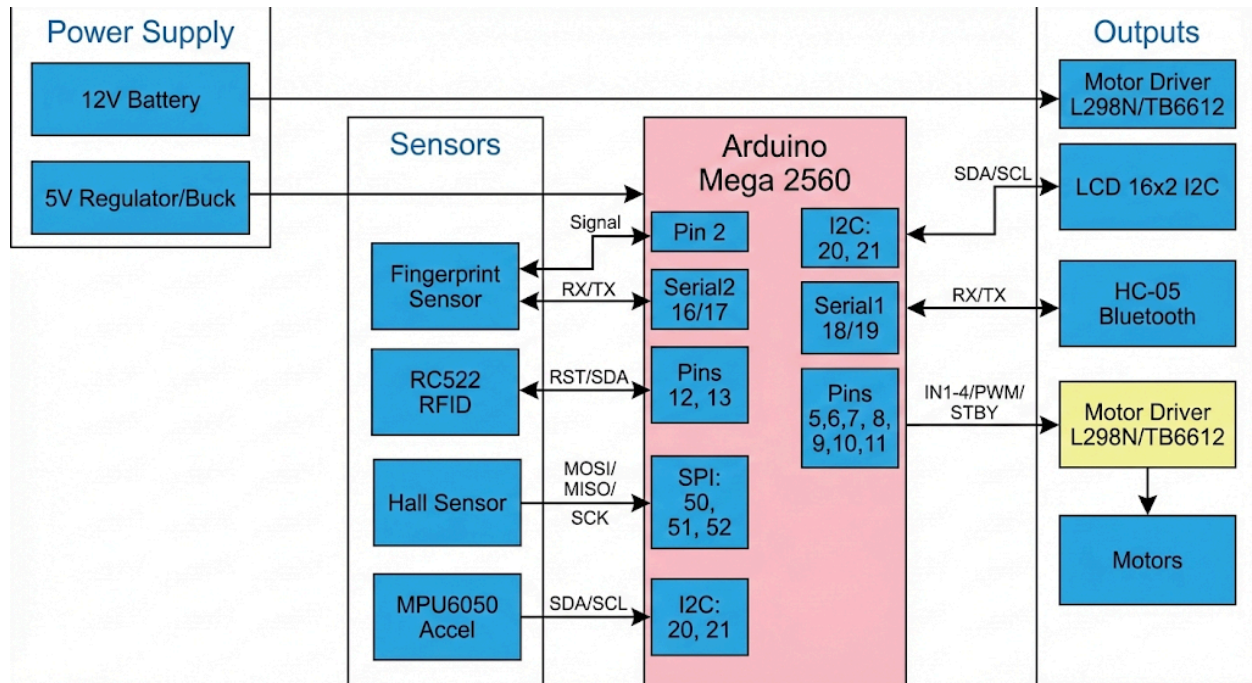
Components & Cost Breakdown:

Components	Quantity	Unit Price	Total
Arduino Mega/Uno	1	1,800.00/-	1,800.00/-
Fingerprint Sensor with UART module	1	2,100.00/-	2,100.00/-
MFRC522 RFID Module	1	175.00/-	175.00/-
MPU6050 Accelerometer/Gyroscope	1	250.00/-	250.00/-
Hall Effect Sensor (Magnetic sensor)	1	60.00/-	60.00/-
TB6612FNG Motor Driver	1	219.00/-	219.00/-
HC-05 Bluetooth Module	1	380.00/-	380.00/-
LCD (16x2 I2C)	1	170.00/-	170.00/-
Jumper Wire Bundle	1	120.00/-	120.00/-
Battery 3.7	3	95.00/-	285.00/-
Chasis	1	400.00/-	400.00/-
N20 Wheels	2	80.00/-	160.00/-
DC motors	2	295.00/-	590.00/-
Magnet	1	25.00/-	25.00/-
Grand Total			6,734.00/-

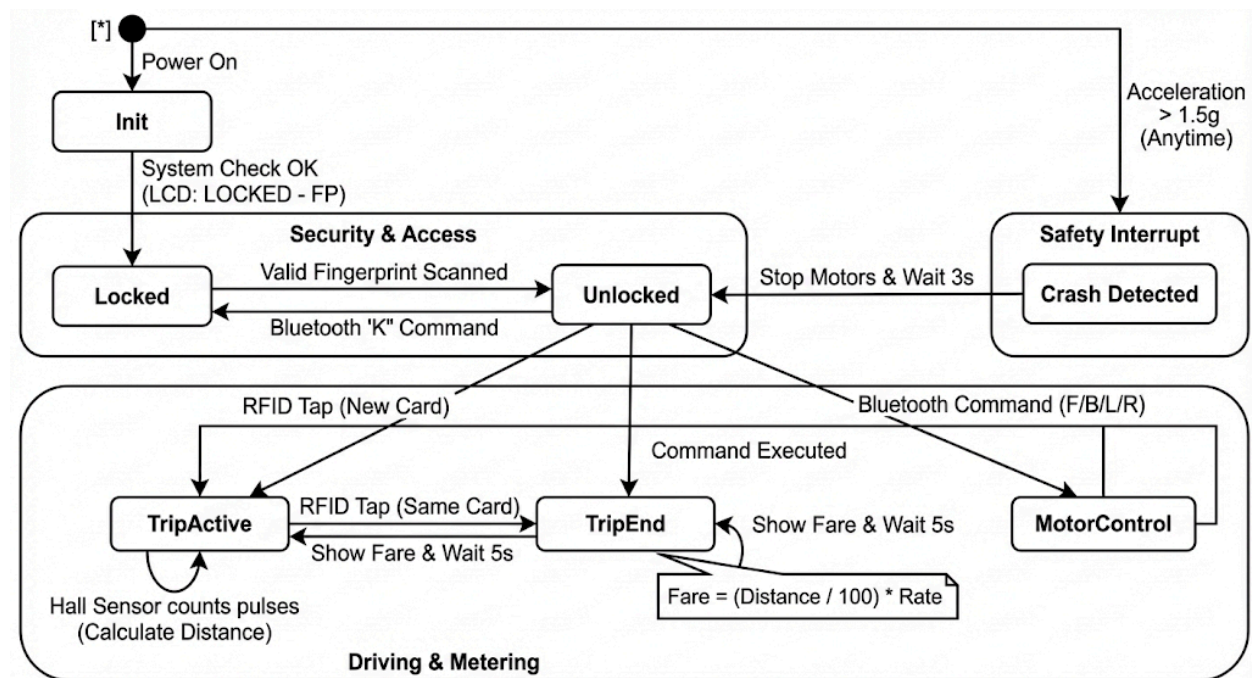
Communication Protocol:

- **UART (Serial):** This is the case with the Fingerprint sensor and the Bluetooth module; since they need asynchronous serial.
- **I2C (Inter-Integrated Circuit):** The LCD and the MPU6050 were designed with the I2C to reduce the amount of wiring necessary, enabling several devices on the SDA/SCL lines.
- **SPI (Serial Peripheral Interface):** This is to be used in the RFID module to handle high-speed data transfer that is needed to read card UIDs.

Circuit Diagram:



Implementation & Workflow:



Challenges & Solutions:

- **Hall sensor:** The Hall sensor indicated more than one pass (bouncing).
- **Fix:** Added a software debounce (millis () - lastPulse > 15) to the Interrupt Service Routine.
- **Problem:** RFID prevented the primary circuit when reading.
- **Solution:** Non-blocking PICCI Checks to verify safety functions (MPU6050) were continuously running.

4. Sustainability & Impact

Sustainability: The system has low power parts and an interrupt based sleep architecture with the display that consumes less battery. The modular design enables some of the sensors (such as the RFID reader) in the system to be replaced without disposing the entire system which encourages electronic reusability.

Impact:

- **Societal:** Guarantees equitable charges to passengers using automated calculation, which eliminates conflict.
- **Industrial:** Gives security to fleet owners (Garage owners) against theft by use of biometric locking.
- **Safety:** The automated crash detection system has a potential of aiding in avoiding serious injuries because it would cut off power when there were erratic movements.
- **Future Work:** We will incorporate the use of GSM (SIM800L) in order to provide the SMS alerts to the owner in case of a crash and improve the power system to include the regenerative braking in order to charge the battery.
- **Limitations:** The existing simulation uses a distance relying on a distance Multiplier that is a Demo because there is not enough lab space. Also, the optical fingerprint sensors can have difficulties with wet or dirty fingers during rainy weather.

5. Results & Discussion

Security: Was able to lock the motors successfully until a registered fingerprint ID was identified.

Fare Logic: The system was correct in determining the fare according to the wheel rotation. As an illustration, a distance of 200m was virtual yielding a fare of 20.00 Tk (10Tk/unit rate).

Crash Test: Vigorous shaking of the sensor (representing >1.5g) activated the "CRASH DETECTED" routine and terminated motors in 300ms.

The findings support the validity of the idea that the implementation of biometrics and accelerators as one control device is possible and effective. The system achieved all the targets as far as security, metering, and safety are concerned.

6. Conclusion

This project was able to prove a prototype of a Smart Rickshaw Meter. Our secure ignition system occurred through fingerprints, our transparent fare system through RFID and Hall sensor, and our proactive safety system through accelerometers. The project emphasizes the importance of retrofitting old vehicles with the IoT technology to sustain and be safe in the transportation industry of the modern city.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Md. Dodi Al Fayed	22301325	Hardware design, Circuit building	• Designed the circuit schematic. • Soldered the motor driver connections. • Assembled the physical chassis.
Jauad Ahmed Sadik	22301342	Software Design	• Wrote the C++ code for Logic and Interrupts. • Implemented the Debounce logic for Hall Sensor. • Integrated the Adafruit Fingerprint library.
Md. Sabbir Akon	22301242	Calibration and Report	• Calibrated the MPU6050 thresholds for crash detection. • Designed the Flowchart and Block diagrams. • Compiled the final report and cost analysis.

References

- Adafruit Industries. (n.d.). *Adafruit Optical Fingerprint Sensor*. Retrieved from <https://www.adafruit.com/product/751>
- LastMinuteEngineers. (n.d.). *In-Depth: Interface MPU6050 Accelerometer & Gyroscope Sensor with Arduino*. Retrieved from <https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/>
- NXP Semiconductors. (2016). *MFRC522 Standard Performance MIFARE and NTAG Frontend Data Sheet*. Retrieved from <https://www.nxp.com/docs/en/data-sheet/MFRC522.pdf>